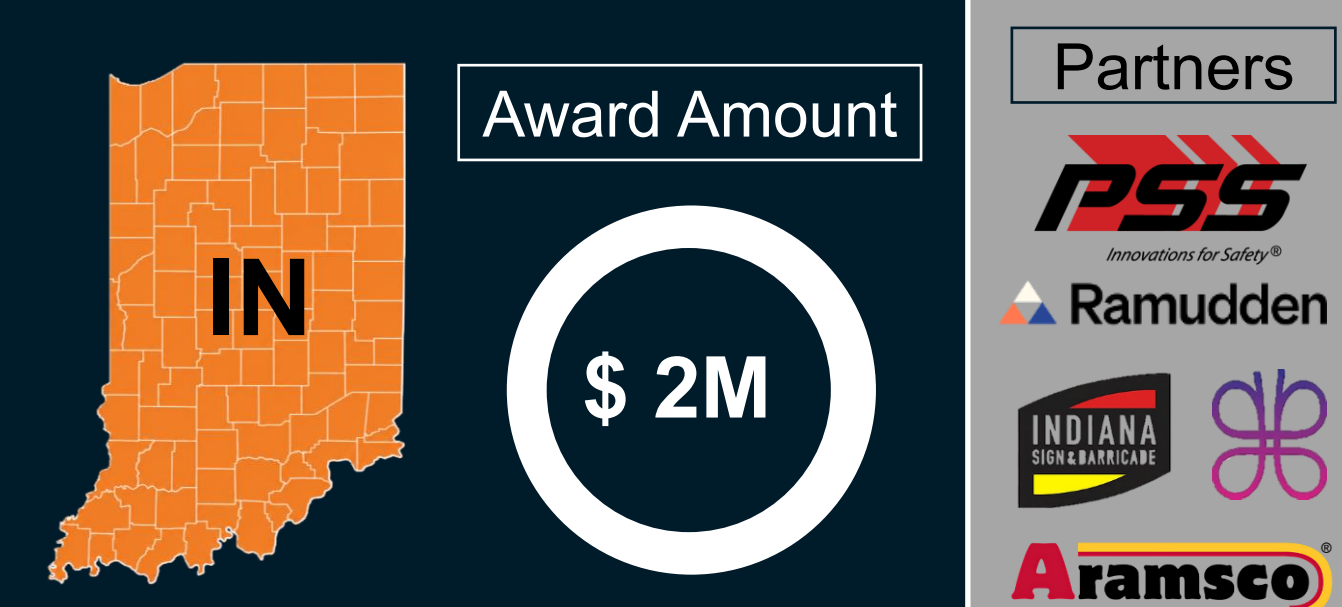


SMART Smart AND Safer Workzone Through Multimodal Intrusion Detection

#69A3552541003



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PROJECT GOALS

- Improve the **safety** and **integration of transportation facilities and systems** for pedestrians, bicyclists, and the **broader traveling public**
- Promote **connectivity** between and among connected vehicles, roadway infrastructure, pedestrians, bicyclists, the public, and transportation systems.

Across the **U.S.** (2023)

Within the **Indiana** (2023)

818 work zone **fatal** crashes

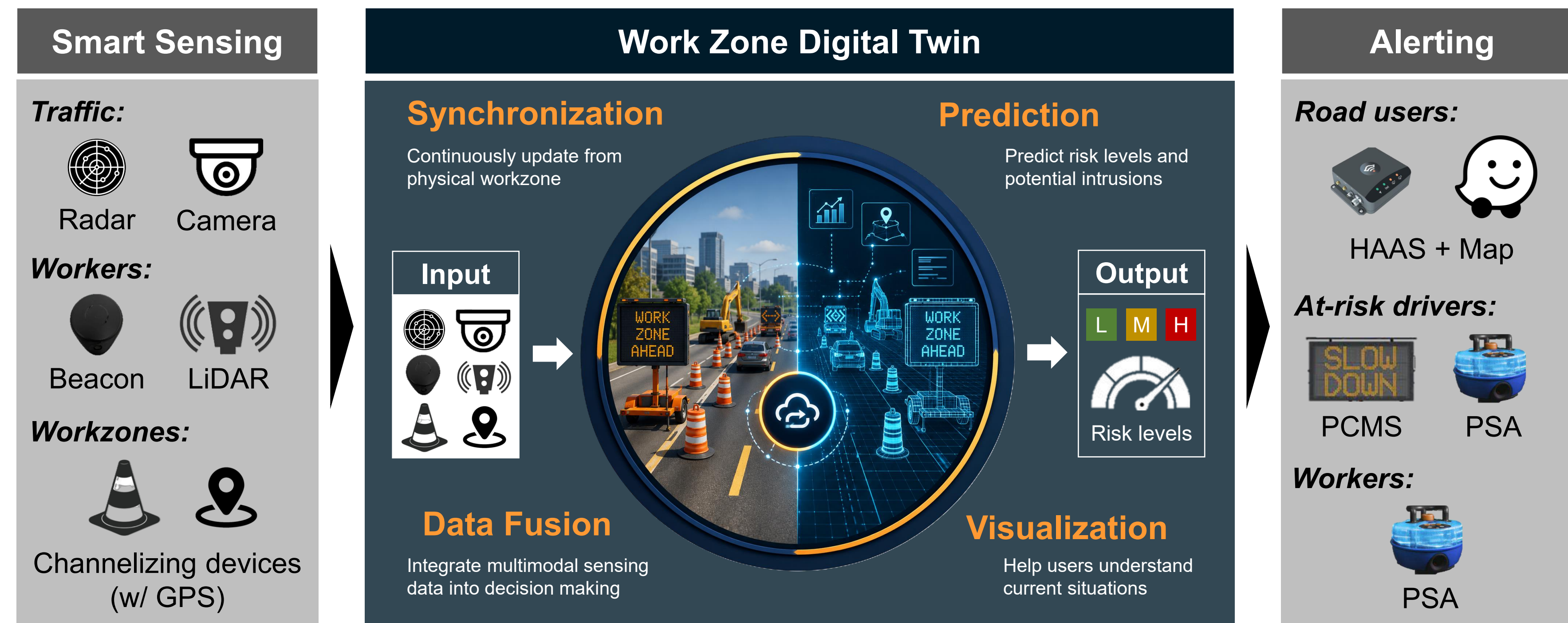
6348 collisions within work zone

899 fatalities caused by crashes

37.7 fatalities **rate** / 1000 collisions

This project aims to develop an integrated **smart work zone system** to address work zone intrusions

DIGITAL TWIN PROTOTYPE



CHALLENGES & LESSON LEARNED

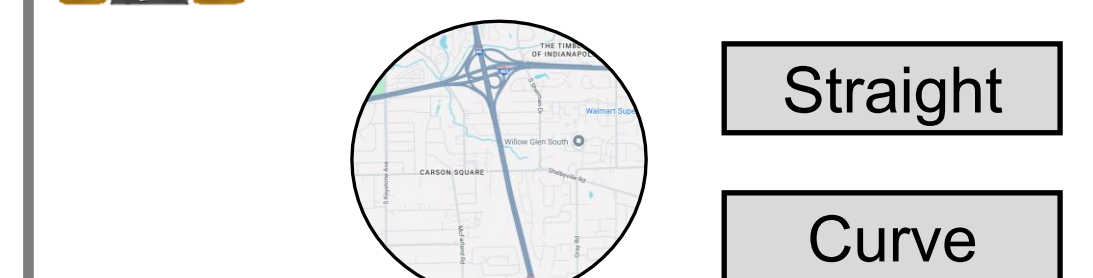
1 Power Supply Limitation in Work Zones

Unavailability of constant & reliable power source



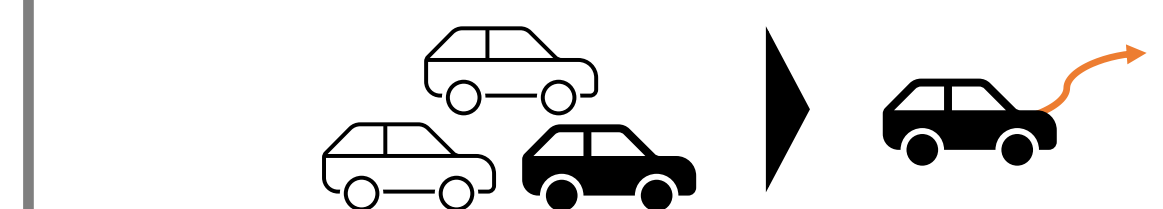
2 Varied Shapes and Scale of Work Zones

Variation & complexity of work zones



3 Computational Efficiency Challenge

AI prediction adds cost and latency



4 Coordination with Contractors

Dynamic schedule & site access for deployment



FIELD DEPLOYMENT

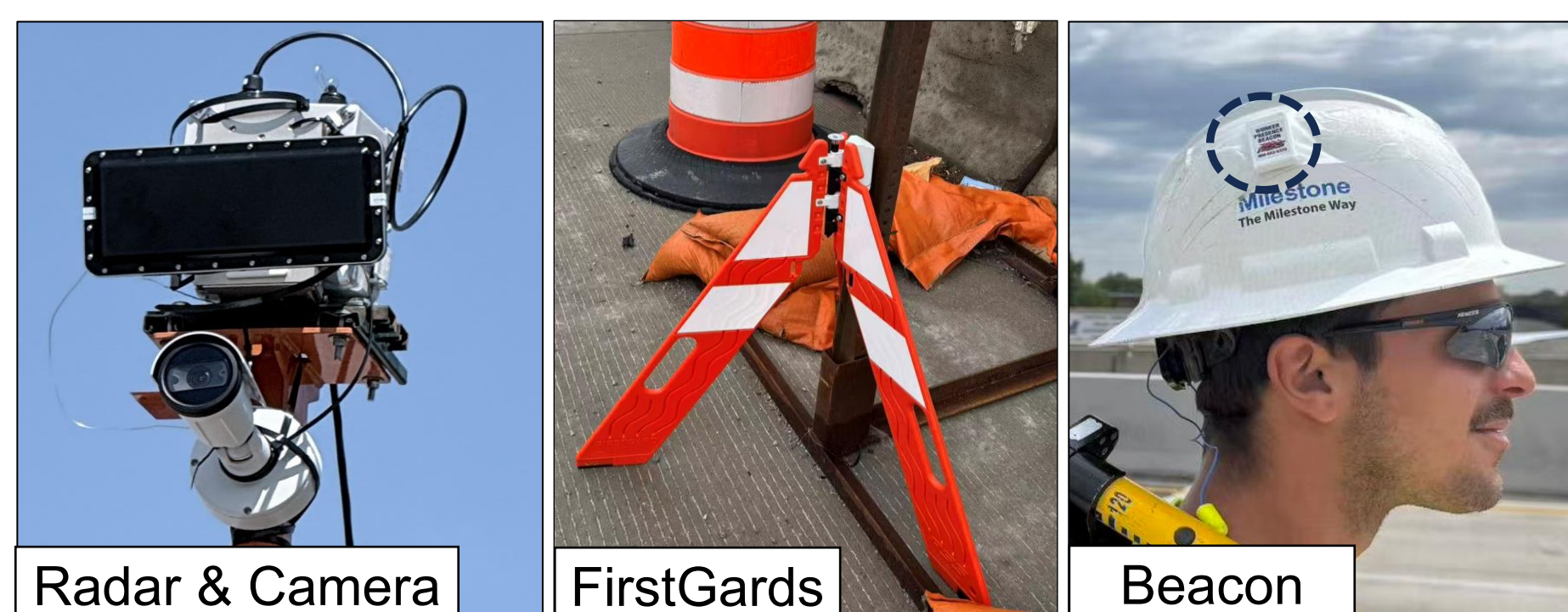
Site1: US31 (Kokomo)



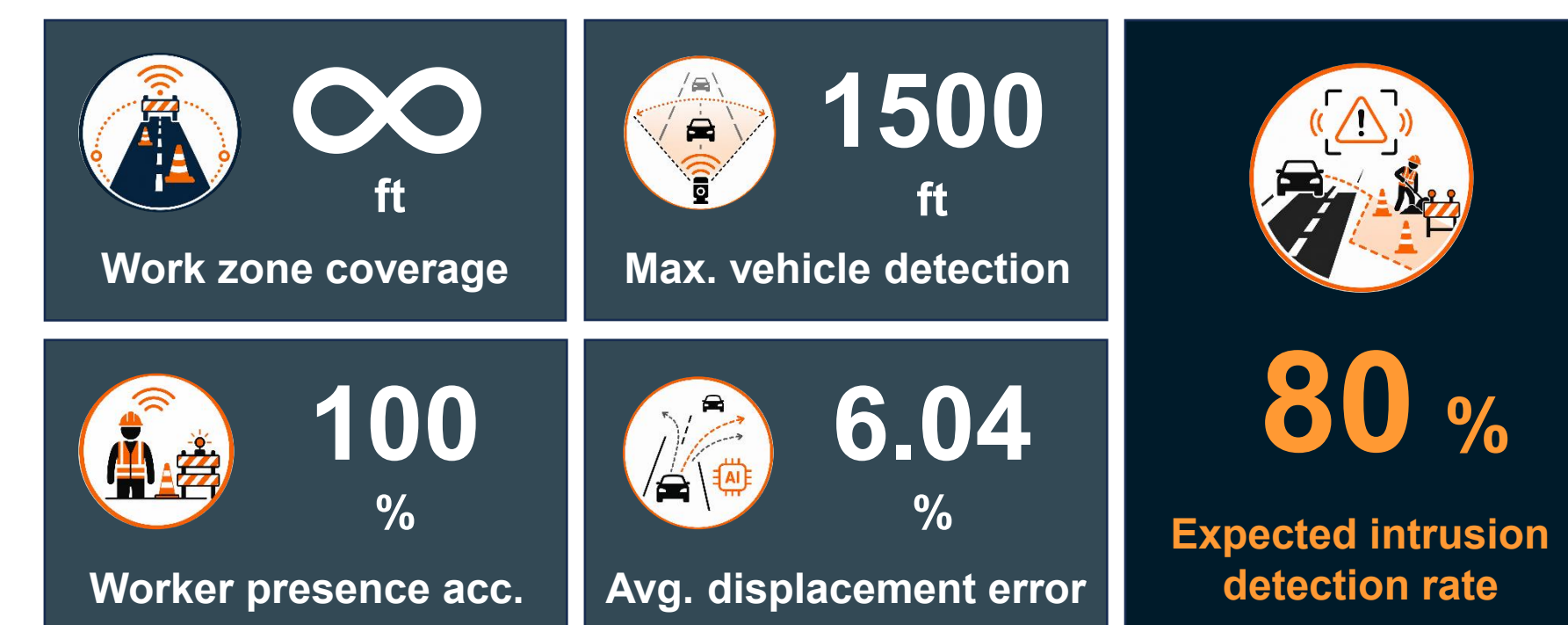
Site2: I-65 (Indianapolis)



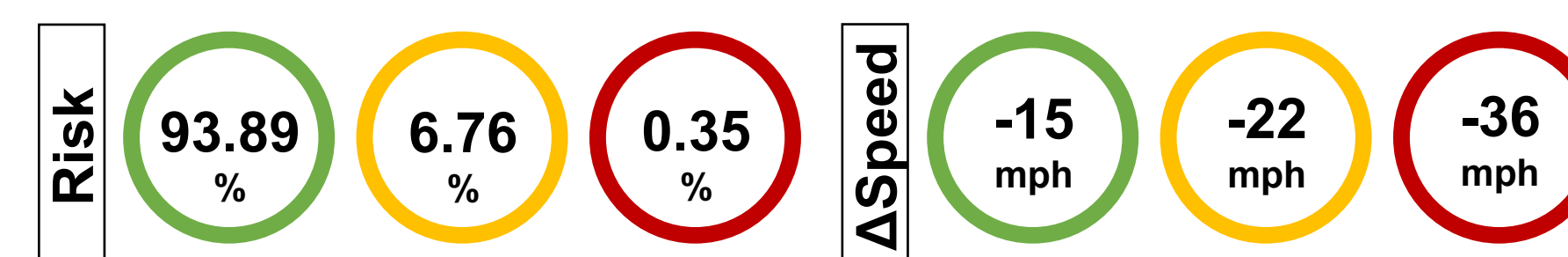
Site3: I-70 (Indianapolis)



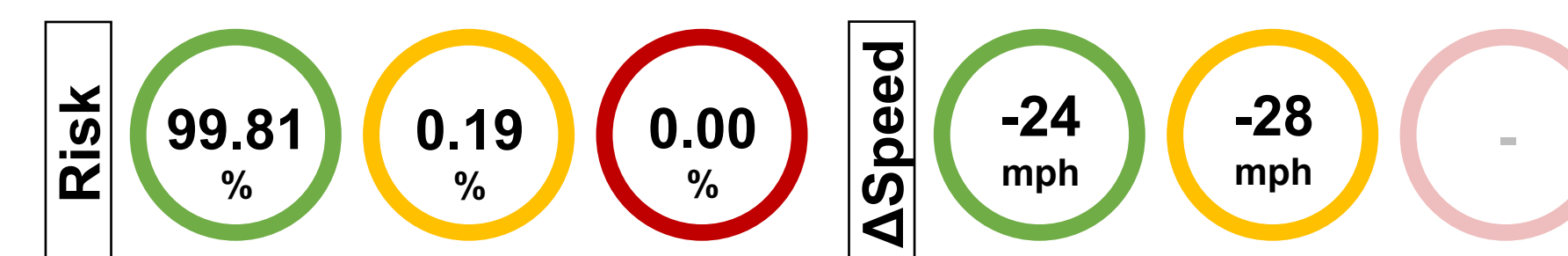
SYSTEM VERIFICATION & VALIDATION



Site1: US31 (Kokomo)



Site2: I-65 (Indianapolis)



Site3: I-70 (Indianapolis)

